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4. Salt tolerant plants (true halophytes) synthesize large amounts of the amino acid proline, galactosyl glycerol and some organic acids which function in osmotic adjustments.

**Mechanism of salt tolerance**

The plants growing in salty habitats like halophytes face the problem of excessive dissolved salts in the solution. Excess of salt creates comparatively more negative osmotic potential so that the plants tend to lose water into surrounding medium. Under such conditions the plants tend to lose water only when their water potential becomes more negative. It is possible only if they absorb excess of salt and accumulate it in their cell saps to maintain the same or higher concentrations as those of outside plants.

**The drawbacks:**

1. Salt accumulates in the vacuoles
2. The plants become succulents
3. Accumulated salt dehydrates the cytoplasm
4. Sodium chloride cannot be tolerated in the cytoplasm and it denatures several enzymes

Thus, absorption and accumulation of inorganic salts fail to solve the problem. The plants however tolerate the salt stress by synthesizing organic compounds that can exist at high salt concentrations without denaturing the enzymes. These organic compounds are called **nontoxic organic osmotica**. Examples: Proline and Betalin (osmoregulators).

## Summary

Growth occurs by cell division, cell elongation and cell maturation. The first phase is lag phase, the second is log phase and the final phase is steady state phase. The log phase is otherwise known as **exponential phase**. The three phases are collectively called Grand period of growth. Plant exhibits plasticity in development. Plant growth and development are controlled by both internal and external factors. The internal factors are chemical substances called Plant Growth Regulators (PGRs). The hormones are classified into five groups: Auxins, gibberellins, cytokinins, abscisic acid and ethylene. These PGRs are synthesized in various parts of the plant. PGRs may act synergistically or antagonistically. The external factors affecting growth includes water, nutrition, temperature, oxygen and light. Mechanism of flowering is controlled by light period (photoperiodism) and temperature (vernalization). The physiological changes on flowering with effect from relative length of light and darkness (photoperiodism) are called photoperiodism. A bluish biliprotein responsible for the perception of light in photophysiological process (induction and inhibition of flowering) is called **Phytochrome**. Besides photoperiod certain plants require a low temperature in the earlier stages for flowering. Many biennial and perennial plants are induced to flower by low temperature ($0^{\circ}C$ to $5^{\circ}C$). This process is called **vernalization** and the reversal effect of vernalization is called **devernalization**. The condition of a seed when it fails to germinate even in suitable environmental condition is called **seed dormancy**. Thus, dormancy can be overcome by following methods such as scarification, impaction, stratification, alternating temperatures and light. Senescence refers to all collective, progressive and deteriorative processes which ultimately lead to complete loss of organization and function. Senescence is of four types and they are overall, top, deciduous